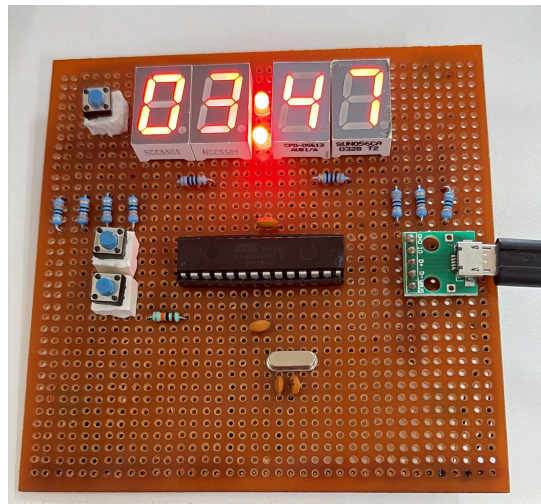

DIGITAL CLOCK

Using ATmega328P Microcontroller

Complete Beginner Friendly Hardware Guide



Microcontroller: ATmega328P-PU

Display: 4 Digit Seven Segment Display

Clock Source: 16 MHz Crystal Oscillator

Implementation: Breadboard, Perfboard & 3D Printed Case

Prepared By

Krishna Janardhan Marri

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1 Introduction

A digital clock is an electronic device that displays time using numerical digits instead of mechanical hands.

The aim of this project is to design a working digital clock from basic electronic components.

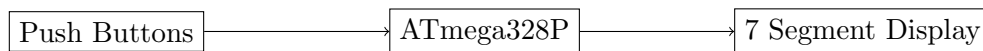
The project starts with an Arduino prototype, then the Arduino board is removed and only the ATmega328P microcontroller is used.

Finally, the circuit is soldered permanently and placed inside a custom 3D printed case.

2 Working Principle

The digital clock works in three main steps:

1. The ATmega328P microcontroller keeps track of time.
2. The program calculates hours and minutes.
3. The output pins control the four digit seven segment display.



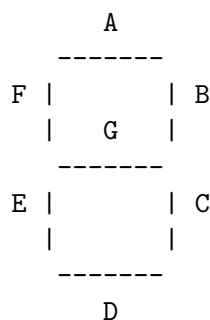
3 Components Required

Component	Purpose
ATmega328P IC	Brain of the clock
16 MHz Crystal	Provides accurate timing
22pF Capacitors	Stabilizes crystal oscillator
Seven Segment Display	Shows time
330Ω Resistors	Protect LEDs
Push Buttons	Adjust hours/minutes
Breadboard	Testing circuit
Perfboard	Permanent circuit

4 Understanding Seven Segment Display

A seven segment display contains seven LEDs arranged in the shape of number 8.

By turning ON different LEDs, numbers from 0 to 9 can be created.



5 Breadboard Prototype

Before soldering, the complete clock was tested on a breadboard.

This helps because connections can easily be changed if mistakes are found.

1. Place the four digit seven segment display on breadboard.
2. Connect segment pins A-G through 330Ω resistors.
3. Connect digit control pins to Arduino output pins.
4. Connect three push buttons:
 - Hour adjustment button
 - Minute adjustment button
 - Start/reset button
5. Upload and test Arduino program.

6 Pin Mapping

Arduino Pin	Connected Component
D2-D8	Display segments A-G
D9-D12	Display digit selection
A0	Hour Button
A1	Minute Button
A2	Start Button

7 Making Arduino Standalone

After testing, the Arduino board is replaced with only the ATmega328P IC.

The IC requires:

- 5V Power Supply
- 16MHz Crystal Oscillator
- Two 22pF Capacitors
- Reset Pull-up Resistor

The bootloader is uploaded using Arduino as ISP.

8 Programming Process

Steps followed:

1. Open Arduino IDE.
2. Upload ArduinoISP example.
3. Connect ATmega328P SPI pins.
4. Select "Burn Bootloader".
5. Upload final clock code.

9 Program Flow

START

|

Setup Pins

|

Read Buttons

|

Calculate Time

|

Display Digits

|

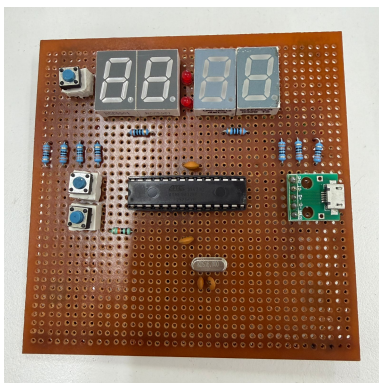
Repeat Forever

10 Permanent Perfboard Assembly

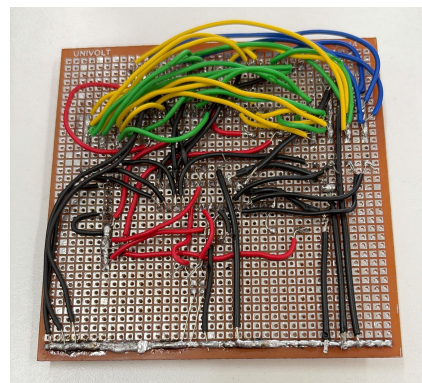
After successful breadboard testing, all components were shifted onto a perfboard.

Care was taken to:

- Avoid short circuits.
- Keep wires organized.
- Check continuity after soldering.



(a) Front View (Component Side)



(b) Back View (Solder Side)

Figure 1: Final Soldered Digital Clock Circuit on Perfboard

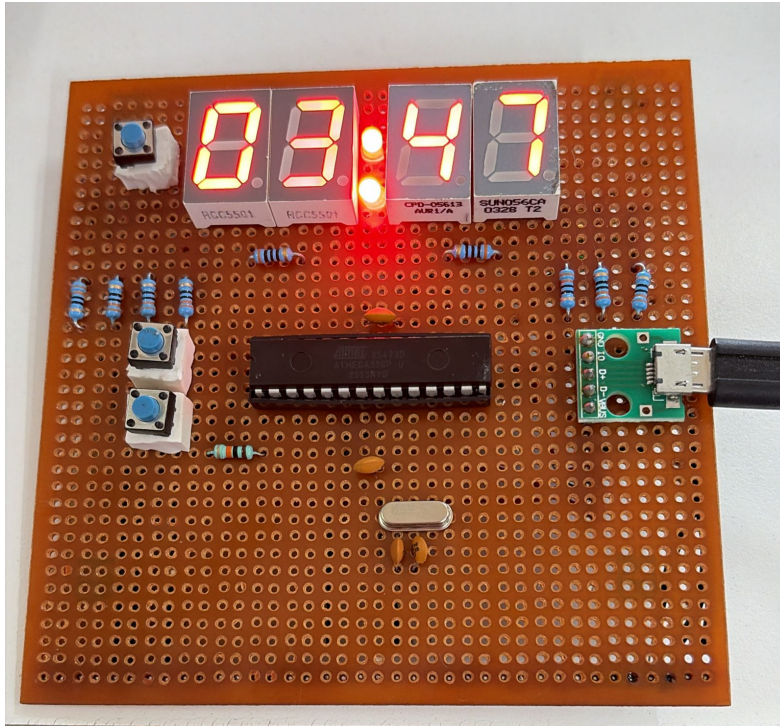


Figure 2: Final Working Digital Clock Displaying Time Successfully

11 3D Printed Enclosure

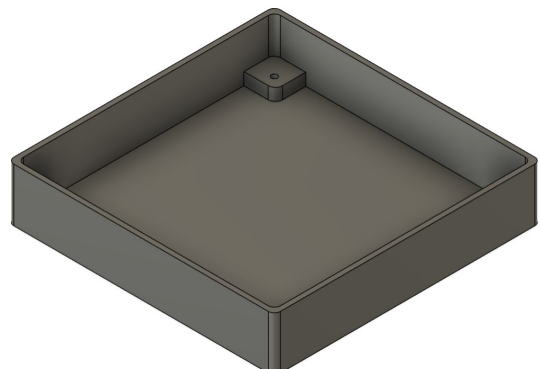
A custom case was designed using CAD software.

Design considerations:

- Display opening.
- Space for circuit board.
- Power cable opening.
- Button access.



(a) Front Case Top View



(b) Case Corner Design

Figure 3: 3D Printed Clock Case Design Views

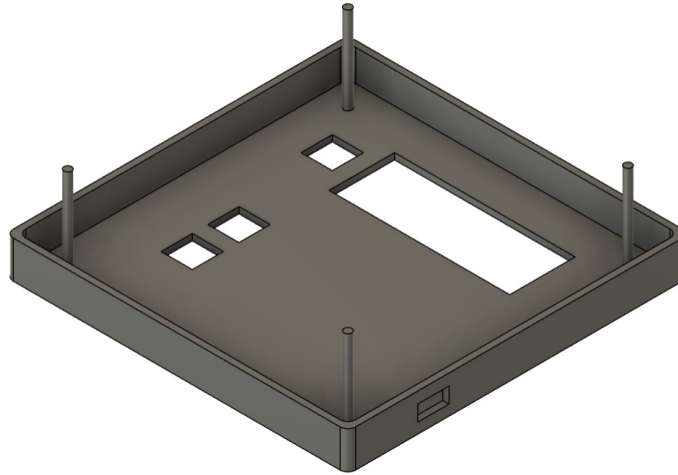


Figure 4: Corner View of the Top Half of Clock Enclosure

12 Troubleshooting

Problem	Possible Solution
Display showing 8888	Check digit pins
No display output	Check power connections
Wrong time counting	Check crystal oscillator
Button not working	Check input wiring

13 Source Code and Repository Access

The complete firmware code, project documentation, images, and CAD design files are maintained using GitHub version control.

The repository contains:

- Arduino source code for programming the ATmega328P microcontroller.
- Hardware implementation images including perfboard assembly.
- CAD files for the 3D printed clock enclosure.
- Complete project documentation.

The complete project repository can be accessed here:

Digital Clock Using ATmega328P GitHub Repository

14 Conclusion

This project demonstrates how a simple Arduino prototype can be converted into a complete embedded system product.

The final clock includes hardware design, programming, soldering, and mechanical enclosure design.